Code No: 123CT **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, February - 2024
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	What are 2's complement and 9' s complement of a numbers? Give examples.	[2]
b)	State and prove De Morgan theorems.	[3]
c)	What is the use of don't care combinations?	[2]
d)	Write the canonical form of the following expression: F (A, B, C, D) = A′ + BC + CD′.	[3]
e)	Draw the logic diagram for 4×2 encoder.	[2]
f)	Can more than one decoder output be activated at one time? Justify your answer?	[3]
g)	What is race around condition? How it is avoided?	[2]
h)	Show the excitation table of JK flip flop.	[3]
i)	What are shift micro operations and what are the different types?	[2]
j)	Draw the memory hierarchy in terms of capacity and access time.	[3]

PART - B

(50 Marks)

2.a) i) Convert 110001.1010010 into hexadecimal. ii) Convert (423.25)10 into Hex.
b) Explain about even and odd parity check with an example, what is the drawback? [5+5]

OR

3.a) Simplify the following Boolean expressions to a minimum number of literals
i) ABC+A'B+ABC' ii) xy + x(wz+wz').
b) Derive the hamming code for the sequence (101011). [5+5]

4.a) Simplify the following Boolean functions, using a four variable Karnaugh map method and implement the simplified function using NAND gates
$F(A,B,C,D) = \sum (0,2,4,5,6,7,8,10,13,15)$.
b) Implement the function $f(a,b,c)=\sum(1,3,4,6)$ using NOR-NOR two level gate structure. [5+5]

OR

5.a) Obtain the simplified expression in product of sums.
$F(A,B,C,D) = \pi(1,3,5,7,13,15)$.
b) Realize the following function Y = A + BCD using NAND gates only. [5+5]

6.a) What is a combinational logic circuit? Implement a Full adder using two half adders and one OR gate.

b) Define a multiplexer? Draw a 4:1 multiplexer for the function $f(a,b,c,d)=\sum(0, 4,5,10,11,12,15)$. [5+5]

OR

7.a) Implement an odd parity generator for 3-bit using a decoder.

b) Explain a binary parallel adder with look ahead carry scheme. [5+5]

8.a) Design a 4-bit universal shift register and explain its operation?

b) Design a mealy type sequence detector to detect a serial input sequence of 101. [5+5]

OR

9. Draw the circuit diagram of a 4 bit UP/DOWN binary counter and explain its working with the help of its state diagram and truth table. [10]

10.a) What is Cache memory? Explain the organization of cache memory with suitable diagrams.

b) Implement the following Boolean functions using PLA with 3 AND gates.
$F1(ABC) = \sum(3,5,7)$, $F2 = \sum(4,5,7)$. [5+5]

OR

11. With a neat diagram, explain in detail about two-dimensional memory decoding scheme. [10]

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